

New Methodology for Surface Potential Decay Measurements: Application to Study Charge Injection Dynamics on Polypropylene Films

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ABSTRACT

Our work shows, through measurements on polypropylene films, the validity of surface potential measurements as a technique for dielectric properties analysis. We propose a representation of the decay and potential return curves that separates different components of dielectric response, by using tdv/dt vs $\log t$ plots. Using this transformation, and varying some experimental conditions we have identified different contributions to the potential decay. From our experimental observations we suggest a qualitative interpretation of the initial seconds of potential decay for negative corona charged samples that uses a model controlled by surface trap emission and charge exchanges between corona discharge and the insulator surface.

Index Terms — Surface potential, potential decay, return potential, charge injection, polypropylene, current measurement, conduction, polarisation, demarcation energy, dielectrics.

1 INTRODUCTION

1.1 SURFACE POTENTIAL MEASUREMENTS

During the last few years, there have been many developments concerning insulator space charge distribution monitoring (see [1, 2] for a review of space charge measurement techniques using the pressure wave). These sophisticated techniques have overshadowed older techniques, among them the characterization of insulators by means of the electrostatic measurement of surface potential of a charged sample (potential decay) or of a neutralised sample (potential return). However, surface potential measurement is a simple and low cost technique and remains still useful in several applications like the study of thin samples for which the new techniques are not useable today. The critical issue concerning potential decay measurements is the interpretation of the curves, since several physical processes (sample polarization,

charge injection, surface conduction, etc.) can lead to similar responses. There exist some mathematical transformations to represent potential decay curves [3, 4, 5] but they usually are linked to a particular assumption about the physical cause of the potential decay.

A general procedure to use surface potential measurements was described by Molinié [6]. However, potential decay measurements must be completed by other kind of measurements as for example current measurements [7] to obtain more information about the dielectric response.

Potential decay in a charged sample with a grounded surface (metal insulator interface) and a free surface (air insulator interface) is a consequence of several physical phenomena which must be understood in order to obtain correct information from measurements. Many physical processes can be involved, the most common are: surface conduction, volume polarization (conduction and dipolar processes), charge injection at interfaces and partial discharges in material voids or irregularities [6]. There is a need of a tool and methodology for separating, as much as possible, every contribution to potential decay.

In our work, we use a general mathematical transformation, the tdV/dt vs $\log t$ diagrams, which can be used for data analysis without considering any previous physical interpretation of the potential decay curves. This has to be done in order to separate phenomena with different characteristic times and amplitudes. Using this direct mathematical operation and varying some experimental conditions (electric field, metallization, temperature, polarization cycles, etc.) we have discriminated between different charge injection contributions to the potential decay.

2 METHODOLOGY

2.1 Mathematical Treatment- tdV/dt Transformation

The immediate $V(t)$ vs t representation usually does not provide much information in itself. The use of a logarithmic time scale is recommended to analyse potential decay results. Concerning the Y-axis, several choices can be made. The derivative form dV/dt vs $\log t$ plot was used by Molinié to show a Curie-Von Schweidler behaviour [3] in epoxy resin. The product of the derivative by the absolute time in seconds (tdV/dt) vs $\log t$ has also been used with some particular interpretations of potential decay measurements (see Watson [5] for the study of charge trapping in polystyrene).

We suggest that the tdV/dt vs $\log t$ transformation is an interesting general tool for potential decay measure-

ments analysis because it allows the identification of a characteristic time and the amplitude of some peaks in the potential decay curves. Figure 1 shows for instance how this transformation allows the identification of several peaks with different amplitudes and characteristic times. This transformation can provide a way to show the differences between very similar curves in form of a peak (Figure 1b).

On a practical point of view, we found it was very convenient to perform measurements at regular time intervals on a logarithmic time scale. The tdV/dt product can then easily be deduced from experimental data, since

$$\frac{\nabla V}{\Delta \log t} \approx \frac{dV}{d \log t} = \ln 10 \frac{dV}{d \ln t} = \ln 10 \frac{tdV}{dt} \quad (1)$$

If we represent some exponential decay curves in the tdV/dt representation, the characteristic time and the amplitude are clearly shown. An exponential decay curve following the equation $V(t) = V_0 e^{-t/\tau}$ appears as a peak with characteristic time τ and amplitude V_0/e (Figure 2). This relation between the tdV/dt curve amplitude and the exponential function amplitude is not valid for other non-exponential functions. Thus, the initial potential value must be a parameter pointed out in the tdV/dt representations.

However, the area under the curve between times t_1 and t_2 is always proportional for any function to the potential loss $\Delta V = V(t_1) - V(t_2)$ as shown by equation (2).

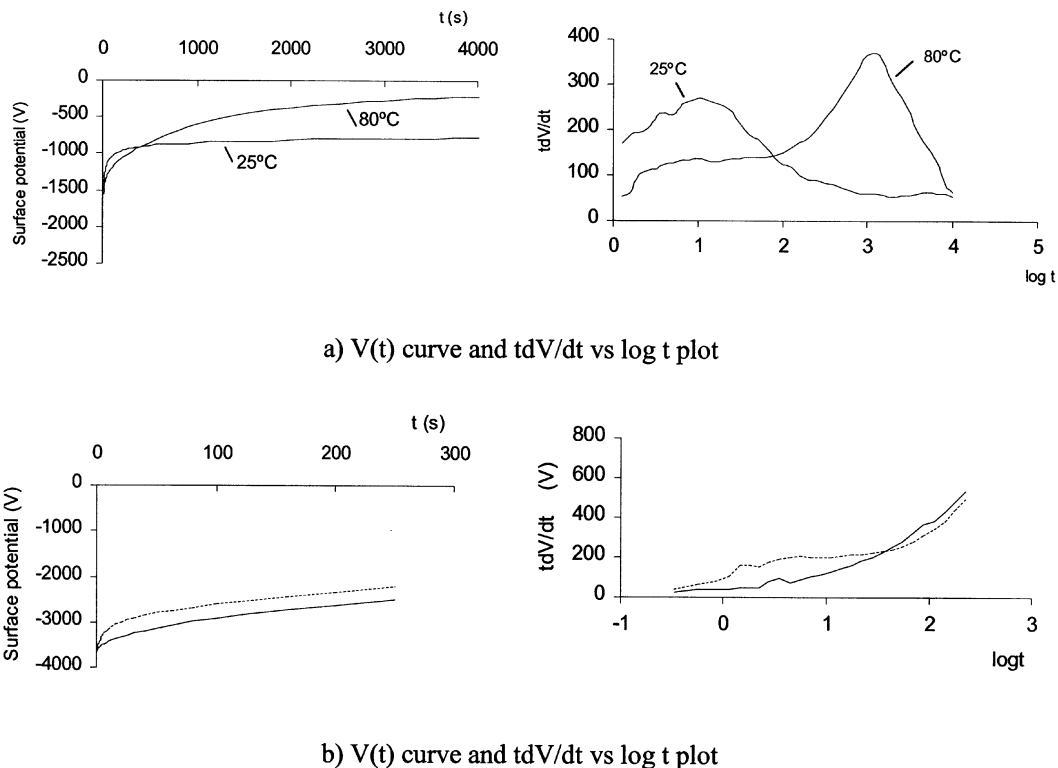


Figure 1. Two examples of the mathematical effect of the tdV/dt vs. $\log t$ transformation for different potential decay curves.

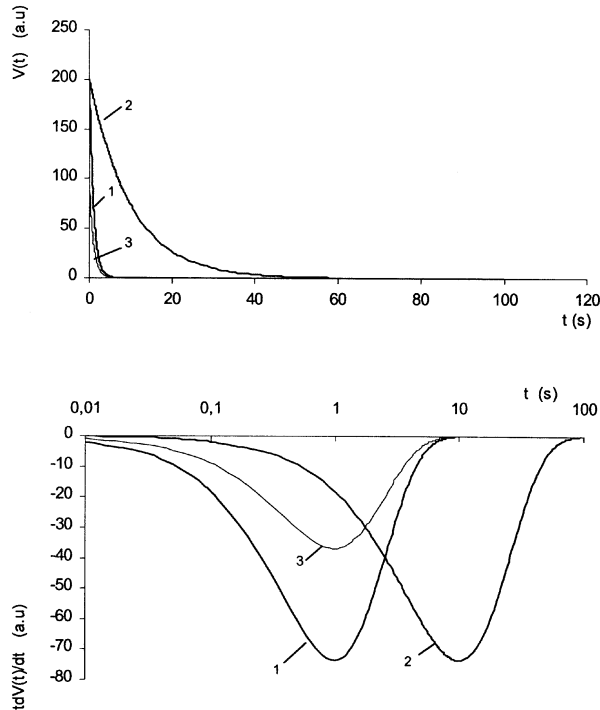


Figure 2. Exponential curves following the expression $V(t) = V_0 e^{-t/\tau}$ in tdV/dt vs $\log t$ plot. 1, $V_0 = 200$, $\tau = 1$; 2, $V_0 = 200$, $\tau = 10$; 3, $V_0 = 100$, $\tau = 1$.

This relationship gives the physical link between the tdV/dt plot and the potential decay.

$$\int_{\log t_1}^{\log t_2} t \frac{dV}{dt} d \log t = \int_{\log t_1}^{\log t_2} \frac{1}{\ln 10} \frac{dV}{d \log t} d \log t = \frac{\Delta V}{\ln 10} \quad (2)$$

The tdV/dt representation can introduce some uncertainty on the characteristic time of two independent phenomena when they overlap in an important period. But two different peaks separated by more than two decades produce small overlapping and can be clearly distinguished.

The tdV/dt transformation is well adapted to potential decay curves because it allows seeing fast phenomena and,

as the potential decay curves tend to slow down for long time and the derivative becomes then small, the product of the derivative and the time permits the representation of the slow phenomena in an appropriated way.

2.2 EXPERIMENTAL METHODOLOGY

The information provided by this mathematical treatment must be completed by an experimental operating procedure to identify the nature of the involved phenomena. Potential decay measurement must be done by varying experimental conditions to show the origin of the phenomena. Some variations on the conditions at the interfaces (metallization) or voltage polarity can show whether phenomena come mostly from the surface or from the volume. The influence of temperature can give information about the nature of involved phenomena for the interpretation of isothermal potential decay.

Potential return measurements [6] can also be used to obtain more information from surface potential decay. Other experiments like current measurements, space charge distribution measurements (if possible) or chemical observations should be used to complement potential decay measurements.

3 EXPERIMENTAL SETUP

We performed potential decay measurements on polypropylene samples cut from 6 μm thick PP film biaxially stretched, most samples having a factory deposited evaporated aluminium electrode whereas a few of them (used for comparison and current measurements) had no metallization. Films are charged on the polymeric side for 2 s under a point corona charger with a grid which allows the control of surface potential (V_0) and discharge energy, whereas the back electrode is grounded (Figure 3). After charging, the sample was transferred under an electrostatic vibrating reed probe to measure surface potential. The transit time between the end of the charging period and the first measured potential is 0.6 s. Surface potential recording is immediately followed by a potential scan on a line segment centered on the potential decay measurement

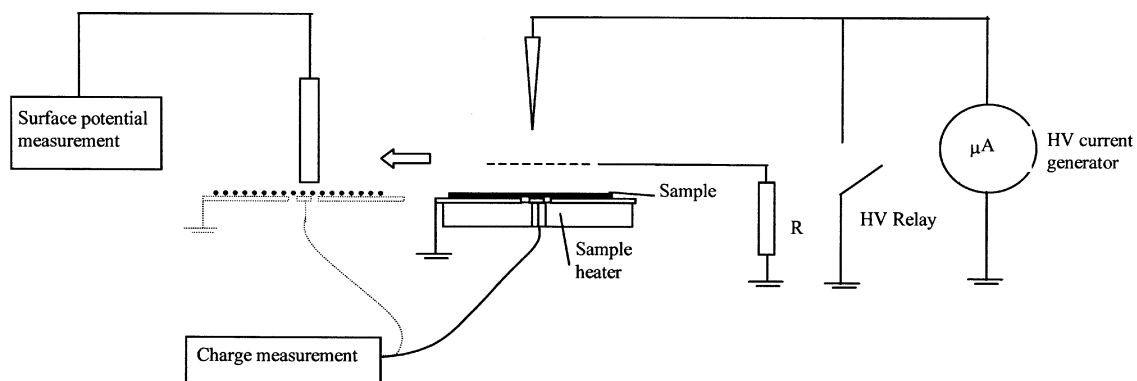


Figure 3. Experimental setup allowing surface potential and charge measurements.

point. Measurements were done in atmospheric air and the temperature was controlled in the 10-80°C range, samples being preconditioned during 6 h at the measurement temperature before the charge deposit. When performing current measurements during corona charging we used a guarded electrode system at the back electrode.

However, contact between the metallic sample holder and non-metallized samples grounded side is only due to the electrostatic forces leading to a non perfect contact. Surface irregularities permit the existence of voids between samples and the electrodes leading to partial discharges inside them. That makes non metallized samples less reproducible than metallized.

The corona discharge arrangement creates a bell-shaped charge deposit on the sample surface as shown in Figure 11 profile 2. There is no lateral diffusion of the charge deposit showing that surface conduction is negligible during potential decay measurements.

Some potential return measurements after a first charge deposit have been done by grounding the grid and applying an opposite voltage to the point discharge, recording then the evolution of surface potential.

4 RESULTS

In our potential decay measurements some reproducible components of the dielectric response of polypropylene can be easily identified in form of characteristic peaks.

4.1 A PEAK

In negative corona charging the main part of the decay (Figures 4, 5 and 6) in the 1-100 s range, corresponds to a peak (A) centred around 10 s. The height of this peak is directly related to charging voltage (V_o) (Figure 4) and corona discharge energy (Figure 6) and it is also very dependant on the measurement position (Figure 5) as shown by the "hollows" appearing in negative polarity on the potential scan data. These hollows are clearly linked to the corona discharge energy transferred on the surface [8], which we can assume to be maximum in the centre of the

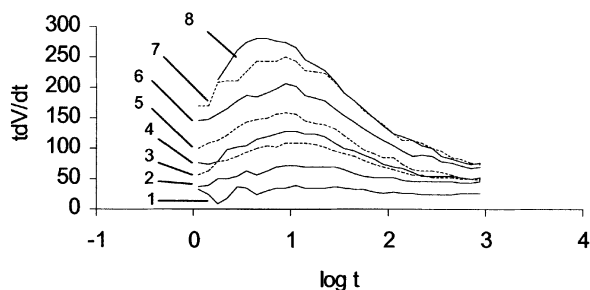


Figure 4. Charging voltage dependence of A peak for a constant corona discharge current ($20\mu\text{A}$) for non metallized samples ($T = 25^\circ\text{C}$). 1, -700V; 2, -1000V; 3, -1500V; 4, -1600V; 5, -1700V; 6, -2000V; 7, -2500V; 8, -3000V.

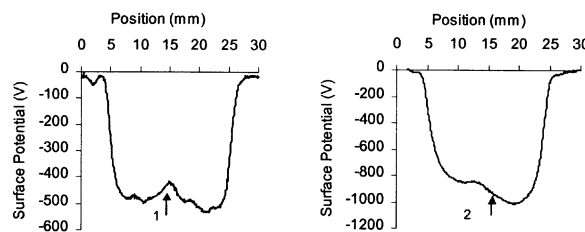
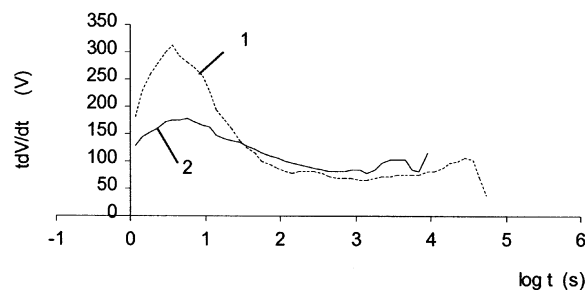


Figure 5. tdV/dt curves for two different samples corona charged ($V_o = -2000\text{V}$ and $T = 35^\circ\text{C}$) and charge profiles after potential decay (the arrow shows the position of the electrostatic probe).

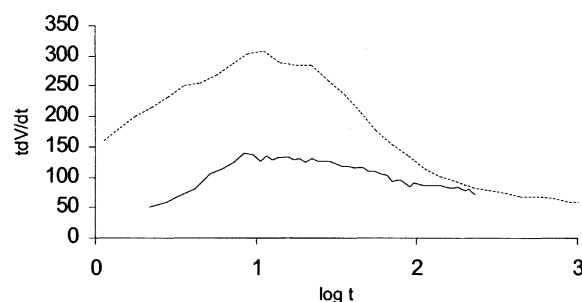


Figure 6. tdV/dt plots for different corona current for samples charged at $V_o = -2000\text{V}$ ($T = 25^\circ\text{C}$). The dotted line is obtained for a current of $40\mu\text{A}$ and the full line for a current of $20\mu\text{A}$.

charge distribution. However, Figure 5 also shows the influence of the film weak points on the dynamics of potential decay at high negative corona charging. Surface profiles after a high negative charge deposit are clearly different from the initial bell-shaped profile. Changes at the back electrode, e.g. the use of non-metallized samples, as well as a silver or aluminium metallization, have little effect on the A peak.

The relation between the height of the peak and temperature is difficult to quantify since other factors, like measurement point, can modify the potential decay dynamics. However, Figure 7 shows how the characteristic time is lowered with rising temperature whereas the amplitudes does not follow such a simple relation (in Figure 7 A peak is maximum at 25°C while A peak amplitude at 40°C is higher than at 35°C).

Above 50°C , our technique is not fast enough to catch the A peak, but in that case, the high potential loss between the charging voltage and the first measured potential value shows its existence.

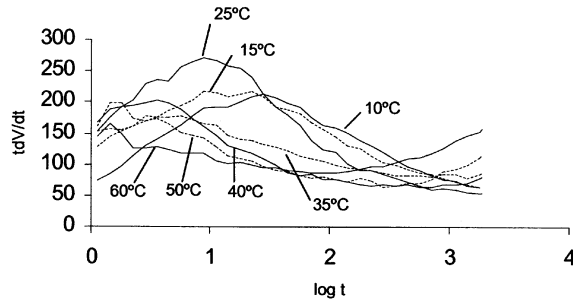


Figure 7. Influence of temperature on A peak ($V_o = -2000$ V, aluminium metallized samples).

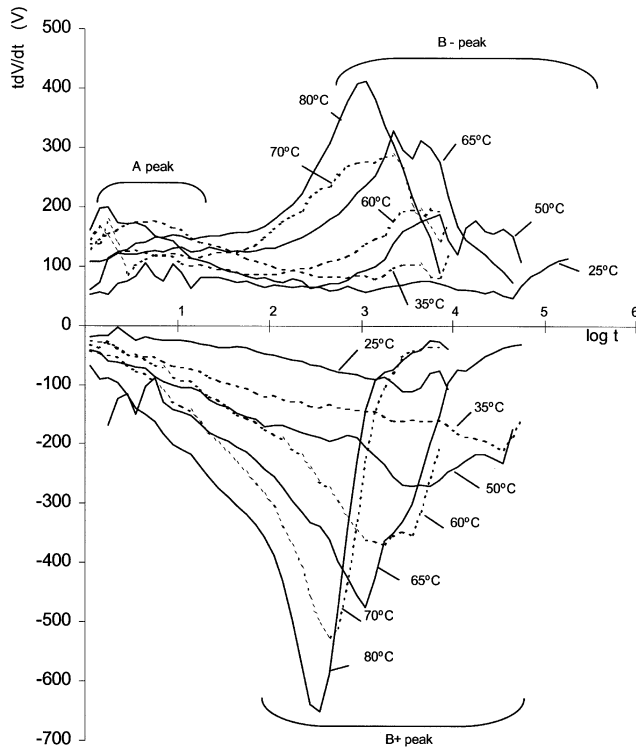


Figure 8. B peak for positive and negative corona charging of aluminium metallized samples ($V_o = -2000V \pm 2000V$).

4.2 B PEAK

When temperature rises, another peak (B) appears in tdV/dt vs $\log t$ plot (Figure 8, for times larger than 10^3 s. This peak is strongly temperature activated, and field dependent (Figure 9). Its shape depends on charge polarity, the amplitude is smaller in the negative than in the positive polarity, and the characteristic time is greater in the negative than in the positive charging.

Potential return measurements (Figure 10) show a higher potential return after a longer previous potential decay and, thus, a smaller voltage step for neutralization.

4.3 C PEAK

For high positive surface potential a wide peak appears (Figure 11) with a characteristic time of 100s. The poten-

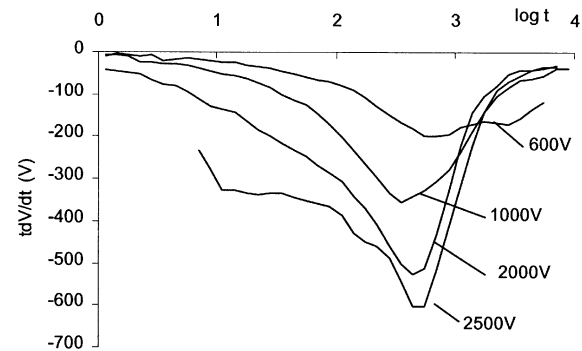


Figure 9. Potential charging influence on peak B+ (aluminium metallized samples, corona charge $V_o = +2000V$, $T = 70^\circ C$).

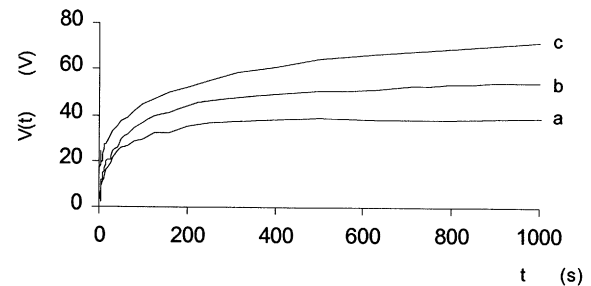


Figure 10. Return voltage after B peak. Positive corona discharge charging $V_o = +2000$ V, $T = 25^\circ C$ and different decay time (t_D) and surface potential before neutralization (V_N). a, $t_D = 100$ s, $V_N = 1405V$; b, $t_D = 300$ s, $V_N = 587V$; c, $t_D = 1000$ s, $V_N = 365V$.

tial scan data show a flat charge profile different from the bell shaped profile corresponding to the corona charging.

5 DISCUSSION

The A peak is clearly related to charge injection at the sample surface assisted by negative corona discharge. However, it should be clarified whether this peak corresponds to a charge transit time inside the sample or a detrapping mechanism characteristic time. That will be discussed in section 5.1.

Concerning the B peak, the existence of a previous A peak in negative charging can explain the amplitude difference between the negative and positive polarity but would not explain the difference in characteristic time which is independent of the initial potential value (Figure 9). So the asymmetry shows clearly the existence of charge injection.

The return potential experiment shown in Figure 10 is also an evidence of charge injection into the material. Two physical causes can theoretically be held responsible for potential return: charge injection and slow dielectric polarization [6]; here dielectric polarization can be ruled out, since in that case, the potential return should be highest in the curve a of Figure 10.

However, charge injection is not enough to explain the potential decay since the return voltages are of small amplitude compared to the corresponding decay. Thus, most

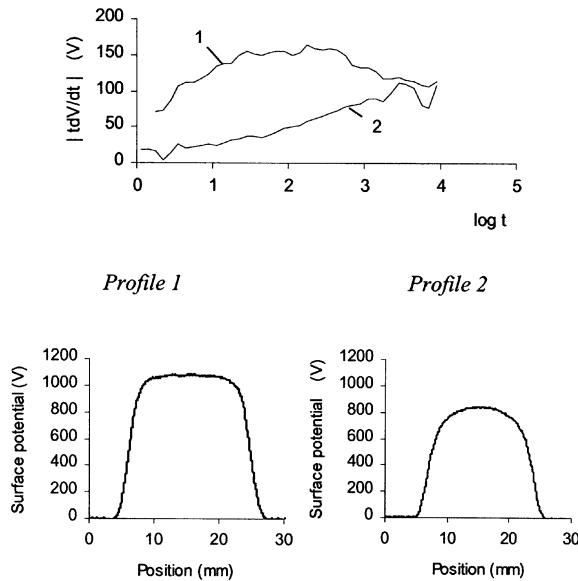


Figure 11. C peak for aluminium metallized samples and comparison to the intrinsic response (charge profiles for curves 1 and 3 were made after potential decay, the profile for electrostatic roll charging is not shown). 1, peak after corona charge $V_o = +2000\text{V}$.

probably, both conduction and charge injection must be considered to explain the B peak. There is probably an intrinsic response in the material (polarization) superimposed on the charge injection. It is known that an ionic conduction is present in polypropylene [9] above 70°C , so we can consider the existence of a temperature activated ionic conduction and an ionic injection of the corona ions into the volume (as shown by Yang [10] on polypropylene films).

As for the C peak, the asymmetry of the response shows that it is not linked to the volume properties. Besides, for non-metallized samples this peak does not appear which points to the interpretation of the C peak as a result of charge injection at the back metallized electrode.

However, it should not be considered that these three peaks only responsible for charge injection; volume polarization will also contribute in all the studied potential decays curves.

5.1 FURTHER DISCUSSION ON A-PEAK MODEL

Charge injection assisted by negative corona discharge is identified as producing the A peak, however a question remains: is the characteristic time of the peak linked to an average transit time of the charge carriers, or is it linked to a typical detrapping energy of the surface traps? Our measurements show the existence of a small average transit time compared to the characteristic time of A peak, thus suggesting a model of the A peak based on progressive surface charge release.

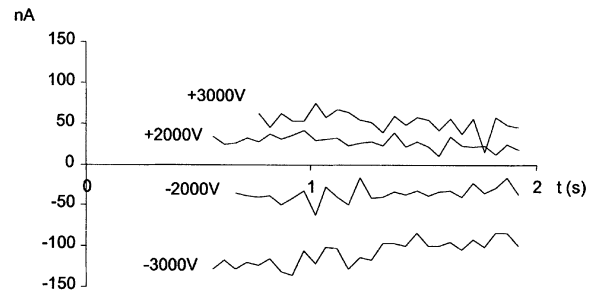


Figure 12. Non capacitive current for non metallized samples (there is a delay of 0.3 s in the measurement due to experimental setup and a delay of 0.3 s also due to calculation of the current).

5.1.1 SMALL TRANSIT TIME

Three items show the existence of a small average transit time or at least, a small transit time for a majority of charge carriers, in our measurements.

1. Assuming an average value of the capacitance of the sample, and showing that the air gap between the charging grid and the sample behaves as a constant resistance, we have calculated the instantaneous surface potential during the charging process. From this, we obtained (using the product of the sample capacitance and the time derivative of the surface potential) a value of the capacitive current, that we subtracted from the total current [11]. This led to the curves given in Figure 12: non-capacitive current under corona charge for non-metallized samples. These curves do not show any rise time or a growing component of non-capacitive current which could be attributed to an average transit time or a distribution of transit times due to distributed charge carriers mobility. The asymmetry of the non-capacitive currents shows charge injection in negative polarity producing a greater current in negative polarity. Besides, the difference between positive and negative current is established in less than 0.3 s, and remains reasonably constant during the measurement that suggests a transit time shorter than 0.3 s.

2. The second evidence for the small transit time is the charging injection made by exposure to neutral excited species of negative corona discharge [8, 12, 13]. When a negative charged sample is exposed to a negative point-grid corona discharge with grounded grid, injection occurs assisted by neutral excited species. But this exposure to neutral excited species does not create a new A peak as shown in Figure 13 whereas the potential loss is 200 V. The charge injected during the exposure to neutral species does not remain in the volume for an appreciable transit time and the potential decay continues with the same dynamics as before the exposure.

3. Finally, the absence of return voltage after A peak shows that there is very few mobile charge remaining in the volume of the sample.

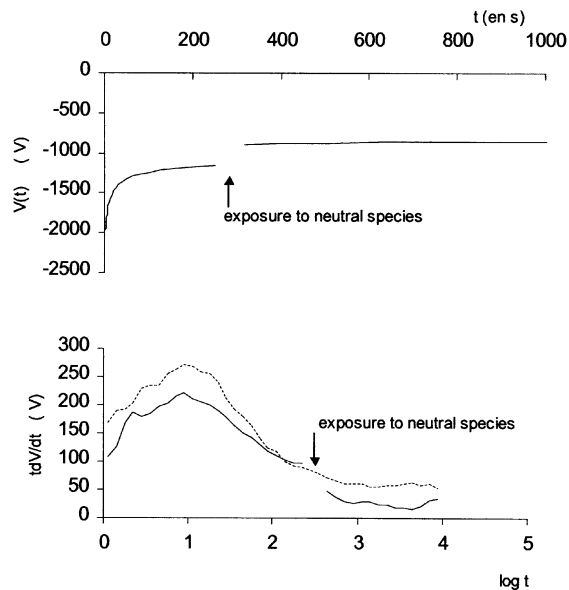


Figure 13. Potential decay and tdV/dt diagram for an aluminium metallized sample charged at -2000 V with a corona discharge $20 \mu\text{A}$ of intensity and exposure to neutral species from a negative corona discharge of $40 \mu\text{A}$, $T = 25^\circ\text{C}$ (The dotted line represents a tdV/dt diagram with no exposure to excited neutral species).

These observations suggest a detrapping dynamics model of potential decay where the limiting step controlling the potential decay rate is the surface traps characteristic release time (a thermodynamic emission model) and not the spatial drift of the charge into the volume (an electrostatic model).

5.1.2 SURFACE DEMARCATION ENERGY MODEL

Simmons and Tam [14] proposed a description of the volume traps detrapping by a demarcation energy model in which the energetic level of empty traps is separated from the level of filled traps by a time dependent energetic level, E_m , called demarcation energy. The demarcation energy level progresses from shallow traps near conduction band to deep traps. Watson [5, 15] applied this model to potential decay measurement to obtain trapping parameters in volume traps for electron beam charged polystyrene. It is easy to prove that the demarcation energy model can be also used for a surface charge detrapping. In our measurements, surface traps are filled by corona charging whereas for electron beam charged samples, the charge penetration in the volume can be experimentally controlled. Moreover, the discussion about transit time shows very few remaining charge in the volume suggesting surface charge as the main fraction of the sample charge.

According to the demarcation energy models, charge decay can be seen as a succession of charge detrapping processes, the beginning of the decay being explained by

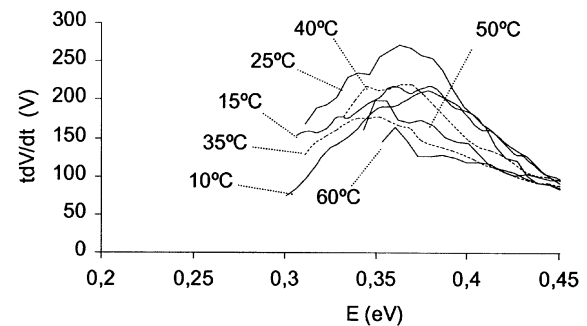


Figure 14. Demarcation energy model applied to curves of Figure 9.

energetically shallow traps emission, the decay on long periods being dominated by deep traps emission. It has been demonstrated [5, 14, 15] that the $tdV/dt = f(\log t)$ plot gives an image of the energy distribution of the traps. Using the demarcation energy model, we can correlate the time values with trapping energies, by the use of the formula: $E_c - E_m = kT \ln(\nu t)$; ν is the frequency, E_c the conduction band energy level and E_m the demarcation energy level. ν is obtained by a statistical calculus using different temperature data [5].

Figure 14 shows the application of this transformation to the tdV/dt diagram of Figure 9 for a temperature range between 10 and 60°C . The A peak is identified with a maximum at $E_c - E_m = 0.36$ eV and an attempt to escape frequency of $\nu = 2 \times 10^5 \text{ s}^{-1}$. That shows that the A peak is a result of a detrapping mechanism of shallow surface traps.

CONCLUSIONS

It is shown that the potential decay measurements can be a powerful tool to characterize the behavior of thin insulating films. Charge injection dynamics and other components of dielectric response can be identified with a combination of a simple mathematical treatment and an experimental procedure. A study of charge injection and trapping in polypropylene thin films using surface potential decay measurements is reported. We proposed a surface demarcation energy model to describe one of the phenomena observed during potential decay. On a practical point of view, we believe that surface potential measurements can become a tool of major importance to follow the charge injection development in thin films, and to detect the first steps of their aging. Further work is being done to lead to an in situ charge injection monitoring apparatus.

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